

GEOGRAPHY

INDIAN GEOGRAPHY



MINERALS & INDUSTRIES

MODULE - 5

- PETROLEUM RESERVES
- PRACTICE Q & A FOR MAINS

INDIAN MINERAL & ENERGY RESOURCES

MINERAL FUELS

PETROLEUM

- 1 Crude petroleum consists of hydrocarbons of liquid and gaseous states varying in chemical composition, colour and specific gravity.
- 2 The word petroleum has been derived from two Latin words Petra (meaning rock) and Oleum (meaning oil).
- 3 Thus petroleum is oil obtained from rocks; **particularly sedimentary rocks of the earth**. Therefore it is called mineral oil.
- 4 Crude petroleum occurs in sedimentary rocks of the tertiary period.
- 5 Oil exploration and production was systematically taken up after the Oil and Natural Gas Commission was set up in 1956.
- 6 Till then, **the Digboi in Assam was the only oil producing region** but the scenario has changed after 1956.
- 7 In recent years, new oil deposits have been found at the extreme western and eastern parts of the country.

INDIAN MINERAL & ENERGY RESOURCES

MINERAL FUELS

PETROLEUM

ON-SHORE OIL PRODUCTION IN INDIA



- Brahmaputra valley of north-east India.
- Barmer area of Rajasthan.
- Gujarat coast in western India.
- Cauvery on-shore basin in Tamil Nadu.
- Andhra Pradesh has both on-shore and offshore oil reserves.

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ON-SHORE OIL PRODUCTION IN INDIA

ASSAM OILFIELDS



- 1 Oldest oil producing state in India.
- 2 The main oil bearing strata extend for a distance of 320 km in **upper Assam** along the **Brahmaputra valley**.
- 3 Oilfields of Assam are relatively **inaccessible** and are distantly located from the main consuming areas.
- 4 Oil from Assam is therefore, refined mostly in the refineries located at **Digboi, Guwahati, Bongaigaon, Barauni**

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ON-SHORE OIL PRODUCTION IN INDIA

GUJARAT OILFIELDS

- Explorations by ONGC have yielded valuable findings of oil bearing rock strata over an area of about 15000 sqkm around the **Gulf of Khambhat**.
- Kachchh, Vadodara, Bharuch, Surat, Ahmedabad, Kheda, Mehsana are the main oil producing regions.
- **Ankleshwar** oilfield is such a prolific oilfield that Nehru called it “**the fountain of prosperity**”.

RAJASTHAN OILFIELDS

- One of the largest inland oil discoveries was made in **Barmer district of Rajasthan**.
- Other important discoveries are **Saraswathi** and **Rajeshwari** oil reserves.

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PETROLEUM

OFF-SHORE PRODUCTION IN INDIA

WESTERN COAST

- Commercial scale of off shore oil production was successful in the regions of **Mumbai High, Bassein and Aliabet**.

EASTERN COAST

- The basin and delta regions of the **Godawari, the Krishna and the Cauvery** rivers hold great potential for oil and gas production.

INDIAN MINERAL & ENERGY RESOURCES

MINERAL FUELS

PETROLEUM

PETROLEUM REFINING



- 1 Oil extracted from the wells is crude oil and contains many impurities. It cannot be used directly.
- 2 It needs to be refined. There are two types of refineries in India:
(a) field based and
(b) market based.
- 3 Digboi is an example of field based and Barauni is an example of market based refinery.

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PETROLEUM

PETROLEUM REFINING



- India's first oil refinery started working way back in 1901 at Digboi in Assam.
- Oil from wells is transported to nearest refineries through pipelines.

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PETROLEUM

PETROLEUM REFINING

ADVANTAGES OF PIPELINE

- Ideal to transport liquids and gases.
- Pipelines can be laid through under water.
- It needs little maintenance.
- Pipelines are safe, accident-free and environmental friendly.

DISADVANTAGES OF PIPELINES

- It is **not flexible**, i.e., it can be used only in a few places.
- Its **capacity cannot be increased once it is laid**.
- It is difficult to make **security** arrangements for pipelines.
- **Detection of leakage** and repair is also difficult.

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MINERAL FUELS

PETROLEUM

UPSC MAINS 2017 GS-1 QUESTION

Petroleum refineries are not necessarily located nearer to crude oil producing areas, particularly in many of the developing countries.
Explain its implications. (15marks , 250 words)



INDIAN MINERAL & ENERGY RESOURCES

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PETROLEUM

POSITIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

- 1 Industrialization & employment generation
- 2 Helps earning foreign exchange
- 3 Avoids price volatility, optimal return to capital invested

NEGATIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

- 1 Causes civil wars and ethnic strifes
- 2 Inter-state/country disputes may arise owing to developmental issues
- 3 Environmental threats

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POSITIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

1

BENEFICIAL TO INDUSTRIALIZATION & EMPLOYMENT



For example :

***Haldia refinery created a variety of
ancillary industries in Kolkata, Hugli etc.***

1

A refinery acts as a “growth pole” around which many industries develop.

2

Refining process generates many useful by-products that has its application in chemical, pharmaceutical, paint and fertilizer industry.

3

This helps in generating employment.

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PETROLEUM

POSITIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

2 HELPS EARNING FOREIGN EXCHANGE



1 Market-based refineries which are **near the ports** can process the crude oil and export by-products to other nations.

2 By this way, we can earn foreign exchange.

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PETROLEUM

POSITIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

3 AVOID PRICE VOLATILITY, OPTIMAL RETURN TO CAPITAL



- 1 Oil fields can get exhausted in the future. Therefore, it is profitable for the refineries to be present near the markets, so that optimal usage of capital invested and logistics can be achieved.

- 2 At times, the oil rich developing countries may be politically unstable, lack skilled manpower and technology to refine the oil.

- 3 Therefore, in a world where refineries were located only near the oil wells, the petroleum prices would be much higher due to
 - Lack of efficiency
 - Lack of economies of scale
 - Frequent political skirmishes.

INDIAN MINERAL & ENERGY RESOURCES

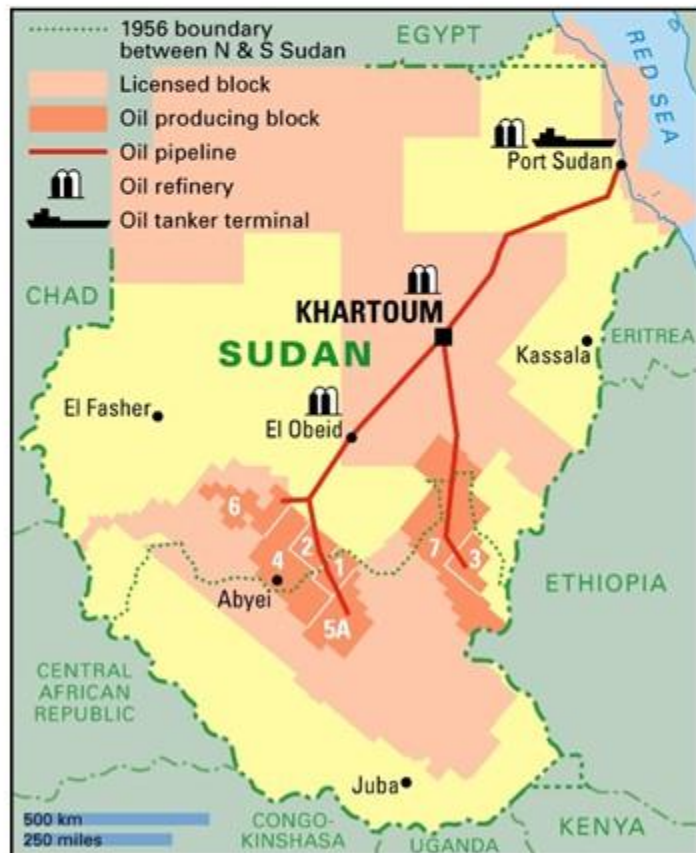
MINERAL FUELS

PETROLEUM

NEGATIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

1

CAUSES CIVIL WARS AND ETHNIC STRIFES



1

Oil is always a politically provoking topic which has quite literally “fuelled” many civil wars and ethnic clashes within countries and across the borders. Here let us take the case study of Sudan.

2

Most of Sudan's oil reserves are in the south, but most of the refineries and shipping facilities were created in the northern Sudan – both because of geographical reasons (easier to export via tankers through Red sea) and political reasons (more jobs for Northern tribes).

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NEGATIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

1

CAUSES CIVIL WARS AND ETHNIC STRIFES



3 This, and many other factors had fueled anger among the south Sudanese that the north had become prosperous using south's natural resources, and yet they were denied adequate share in governance of Sudan.

4 After decades of ethnic strife and civil war, South Sudan gained independence from Sudan (in 2011.)

5 But, South Sudan is a **landlocked country**, and its only oil-pipeline to the sea runs through (North) Sudan.

6 Thus, **South Sudan's oil reserves are useless without the north's cooperation**, which has created further acrimony in their bi-lateral relations.

NEGATIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

2

INTER – STATE DISPUTES MAY ARISE
OWING TO DEVELOPMENTAL ISSUES



1

Post-independence, the first major discovery of oilfields in India was in Assam in the 50s.

2

Industrialists were of opinion that refinery should be located at Calcutta, because then the refinery could sell the products in and around Calcutta.

3

Assamese leaders wanted the refinery to be setup within their state to create jobs & economic opportunities.

NEGATIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

2

INTER – STATE DISPUTES MAY ARISE
OWING TO DEVELOPMENTAL ISSUES

However, Government of India decided to setup the refinery at Barauni in Bihar because:

- 4 For **defense and strategic reasons**, opening refinery at Assam was unsafe due to possible attacks from China and Eastern Pakistan.
- 5 **Barauni** refinery's products could be distributed in the upcountry region via Patna-Nagpur- Amritsar, using the **favourable and faster transportation system** than the products arriving from other ports like Bombay.
- 6 This would also force many port-refineries to cut down their prices.
- 7 According to some critiques, Congress's inner-party politics was more in **favour of creating jobs in Bihar than Assam**.
- 8 But it led to widespread protests and hartals in Assam in the 50s and 60s. Ultimately, PM Nehru was forced to setup two refineries, one at **Noonmati** near Guwahati in Assam and the other at **Barauni** in Bihar under the public sector.

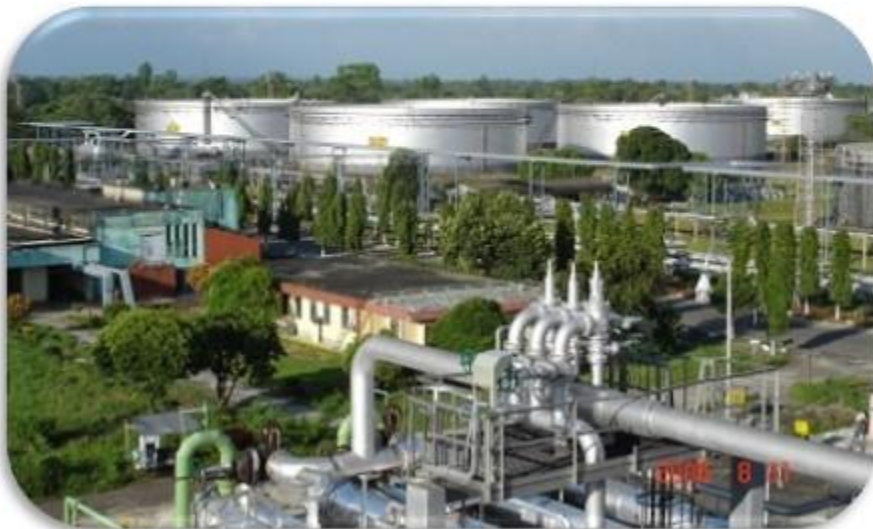
NEGATIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

2

INTER – STATE DISPUTES MAY ARISE
OWING TO DEVELOPMENTAL ISSUES

9

However, over the years, both locations have proved to be unsuitable for oil refining because:



10

Noonmati refinery is located near Bramhaputra to draw its water for its refining processes and for its township. But, the silting in Bramahputra creates water-crisis on routine basis, which

- forces this refinery to partially shutdown its operations, and
 - begin dredging & de-silting operations.
- Thus it generates less than optimal profits.

11

Barauni Refinery was ordered to temporarily shut down by the Union government (2018) because of grave contamination of Ganga river.

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NEGATIVE IMPLICATIONS OF REFINERIES AWAY FROM OIL WELLS

3

ENVIRONMENTAL THREATS

- 1 Refineries transport the crude oil via pipelines. This may lead to **oil spillage problems**.



- 2 Refineries emit **harmful gases**. Terrorist organisation consider them as **high value targets**, therefore, refineries closer to markets pose threat to both environment and human habitats.

- 3 Example:
 - Mathura Refinery – threat to Taj Mahal
 - Barauni Refinery – threat to River Ganga.

ENERGY RESOURCES

NATURAL GAS



- 1 The **Gas Authority of India Limited** was set up in 1984 as a public sector undertaking to transport and market natural gas.
- 2 It is obtained along with oil in all the oil fields but **exclusive reserves** have been located along the eastern coast (Tamil Nadu, Odisha and Andhra Pradesh), as well as Tripura, Rajasthan and off-shore wells in Gujarat and Maharashtra.

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